Ray Tracing in Entertainment Industry

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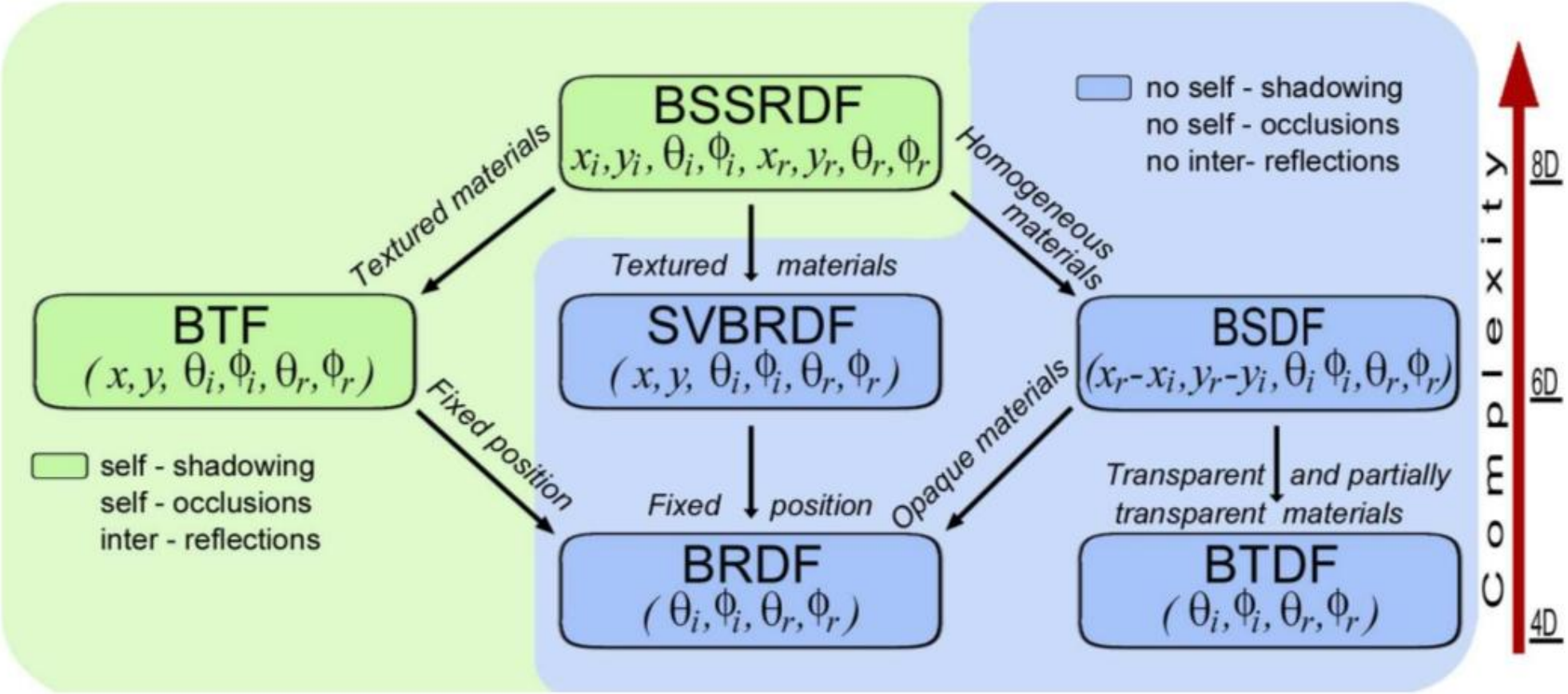
Week 13
Reflectance measurement

How to digitize real-world materials ?

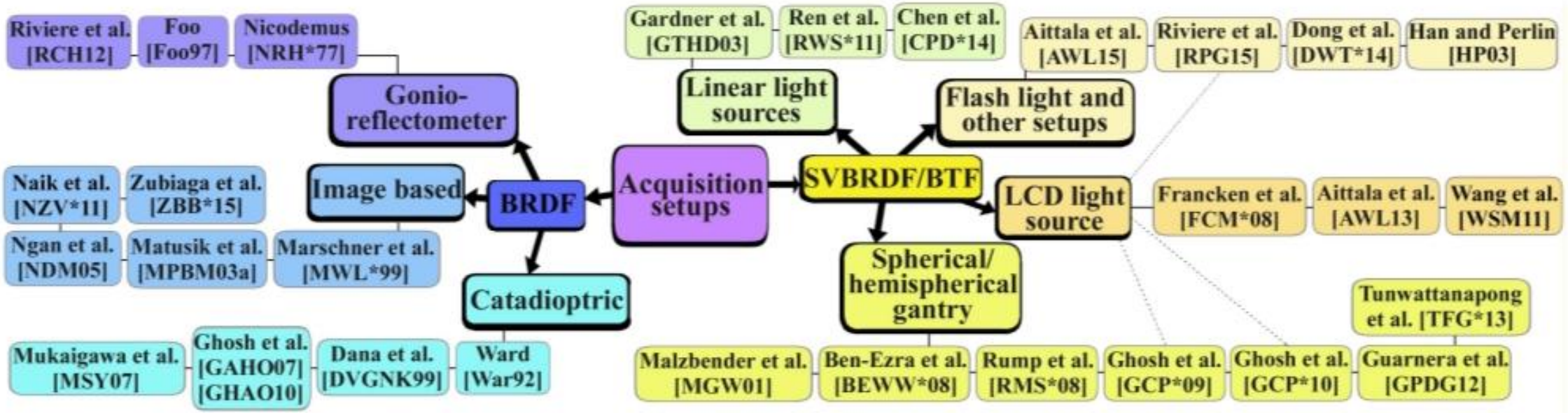
- Terminology
  - Material, Reflectance acquisition
  - Material, BRDF, BXDF scanner
  - Gonioreflectometer
  - Image-based acquisition system
  - Catadioptric

https://eprints.whiterose.ac.uk/161383/1/BRDF_representation_and_acquisition_preprint.pdf

BXDF taxonomy



Acquisition setup



BRDF

$$f_r(\mathbf{v_i}, \mathbf{v_r}) = \frac{dL_r(\mathbf{v_r})}{dE_i(\mathbf{v_i})} = \frac{dL_r(\mathbf{v_r})}{L_i(\mathbf{v_i})\cos\theta_i d\omega_i}$$

Acquiring Parameters for Rendering

http://web.cs.wpi.edu/~emmanuel/courses/cs563/write_ups/zackw_measurement.html

Storing measurement

Uniform and Non-uniform grid - This naive approach stores the sample points directly as a grid. To obtain the BRDF values between sample points neighboring sample points much be interpolated. One approach uses quadrilinear interpolation in which the 4 neighboring sample points are interpolated. This approach doesn't provide tangential continuity. Also, interpolation for non-uniform grids are more difficult.

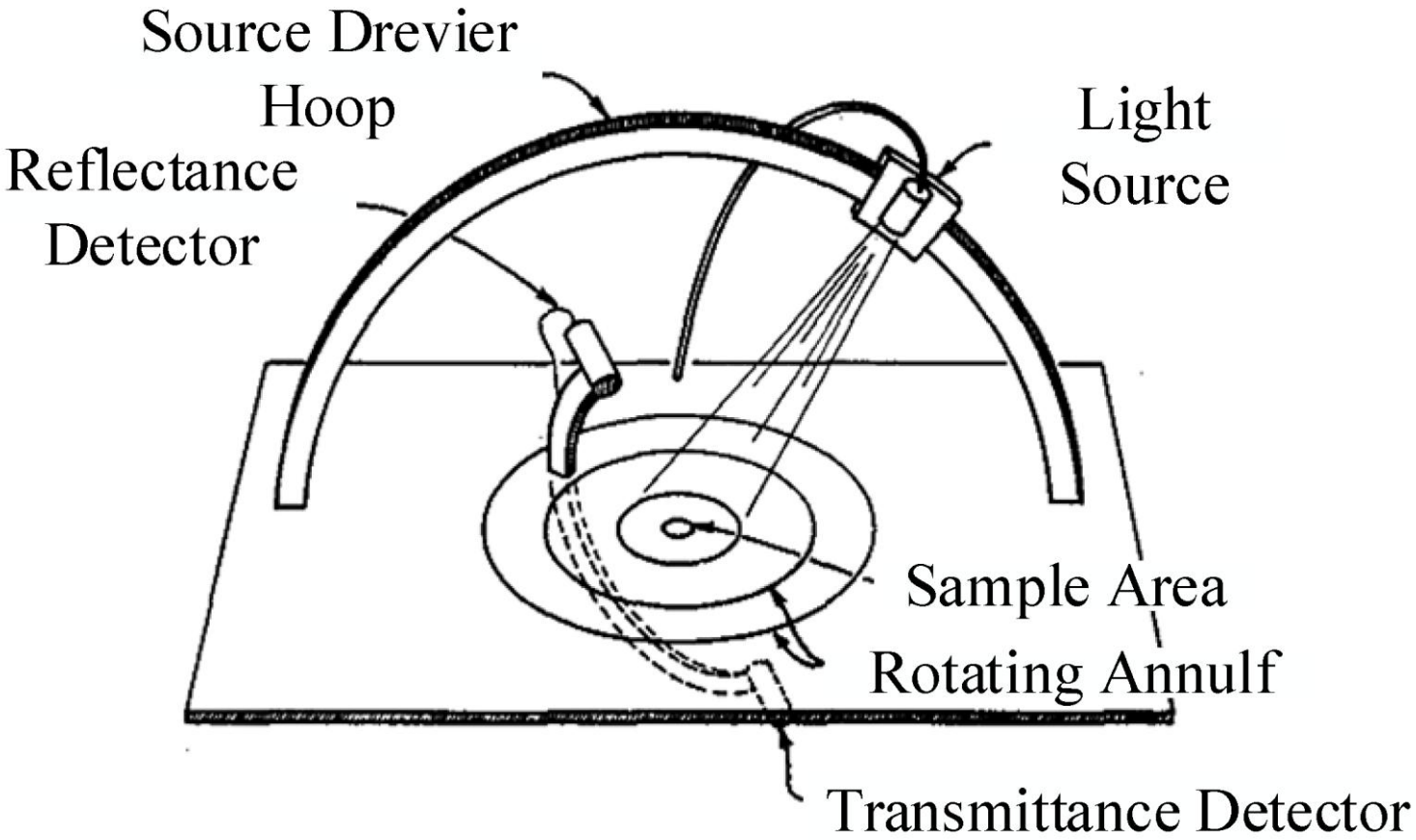
Interpolating Spline - Fits an interpolating spline through the sample points. This approach provides smooth interpolation between sample points but often introduces high frequency noise due to oscillations in the curve. The noise shows up as artifacts in rendering.

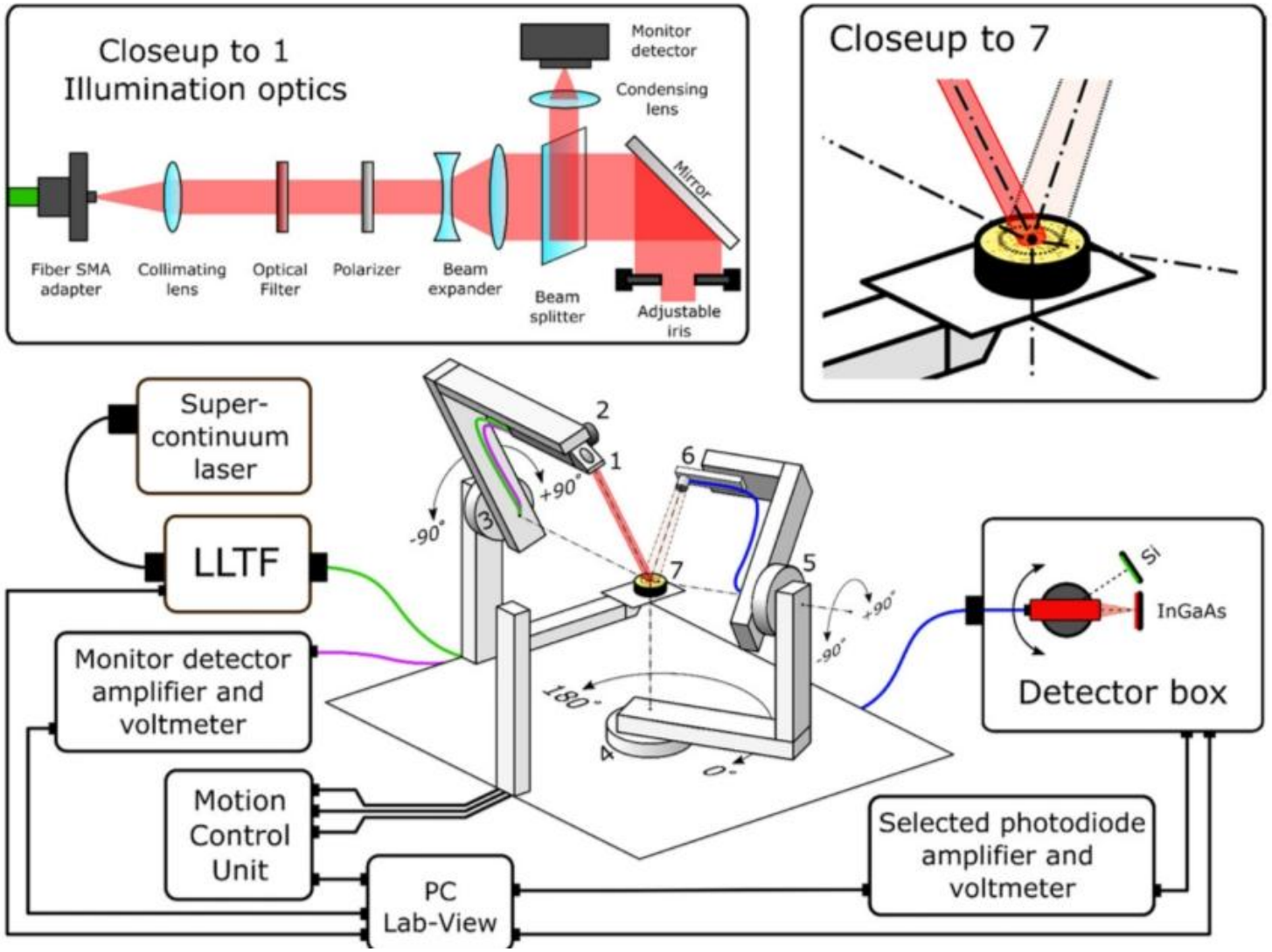
B-Spline - Uses the sample points as control points for the spline. It provides smooth interpolation but the BRDF values at the sample points are not the original measured values.

Spherical Harmonics - This common technique represents BRDFs as a infinite sum of spherical harmonic basis functions. Large number of terms are needed to accurately represent general BRDFs and this technique also introduces high frequency noise. There are several techniques for noise reduction that are used.

Wavelets - Wavelets are basis functions that are good at representing signals that contain a great deal of high frequency signals in some areas while little in others. Wavelets are very attractive for storing BRDFs for this reason. Wavelets also provide good compression while introducing little error.

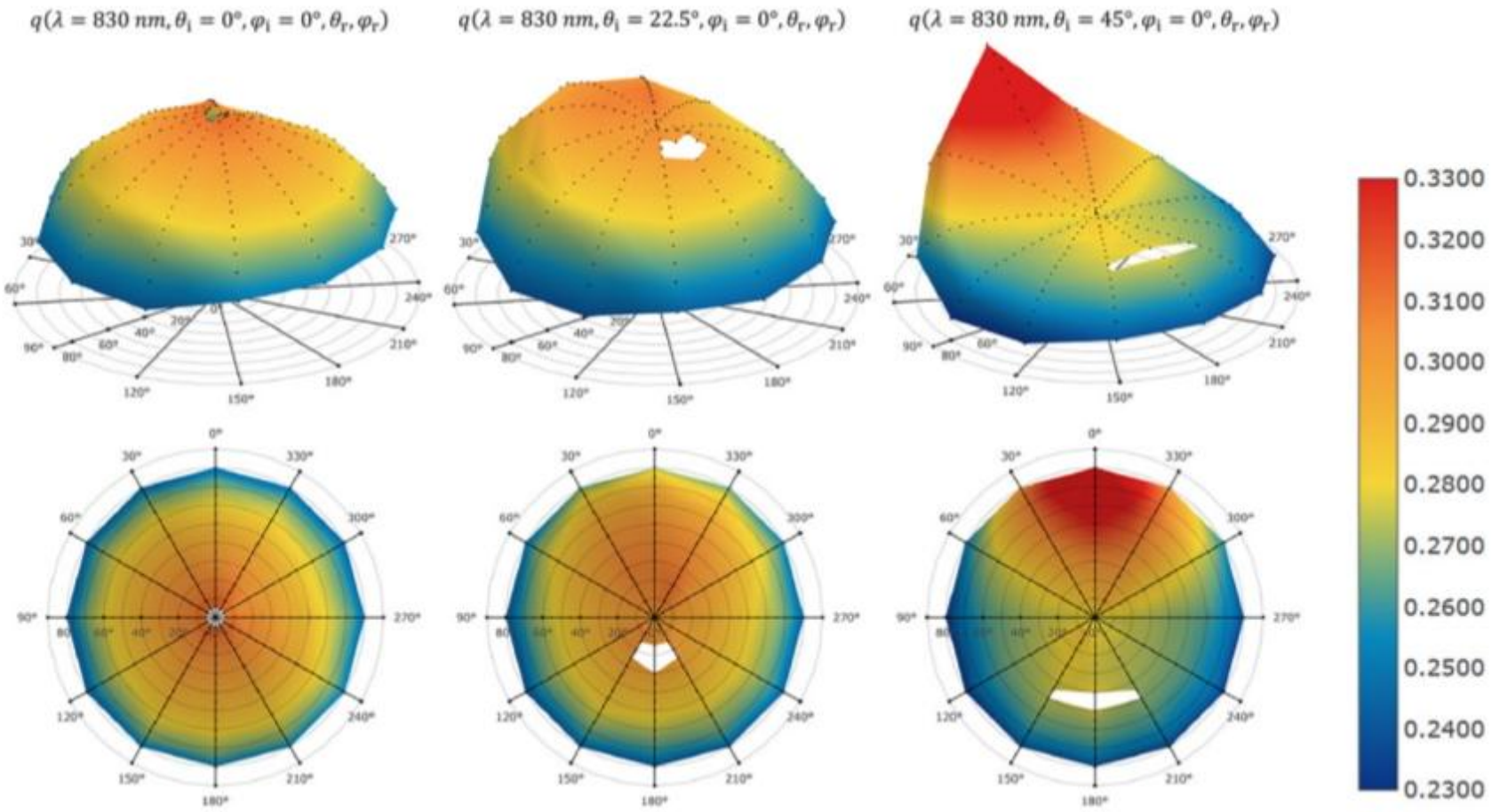
Gonio-reflectometer



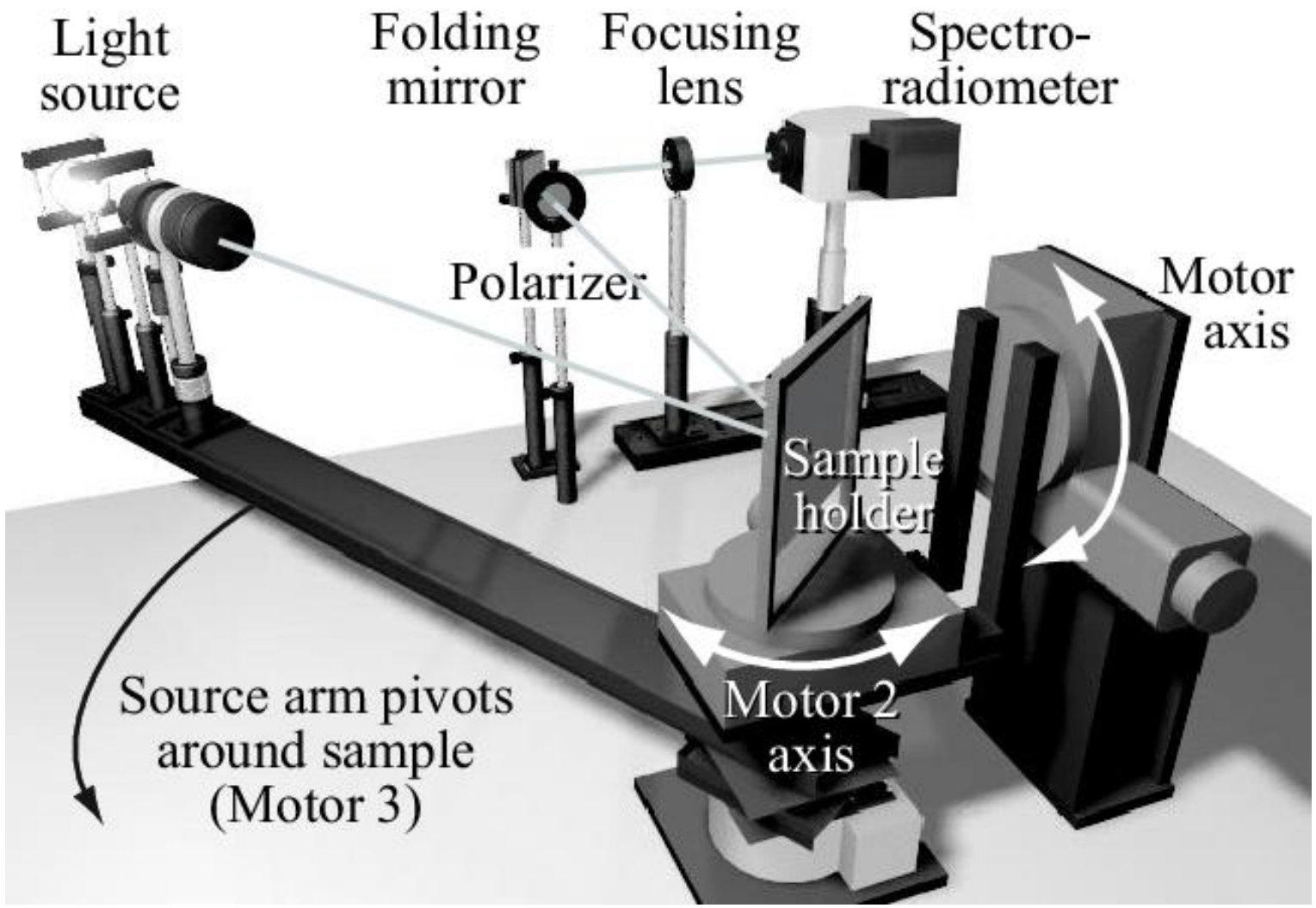


Illustrations of a gonio-reflectometer

Visualization of the examples



A gonio-reflectometer



Spectro radiometer spec

Measurement type			
Ground truthing	✓	✓	✓
Remote sensing	✓	✓	✓
Technology			
Near-infrared Spectroscopy (NIR)	✓	✓	✓
Spectral resolution	3 nm @ 700 nm, 6 nm @ 1400 nm, 6 nm @ 2100 nm	3 nm @ 700 nm, 8 nm @ 1400/2100 nm	3 nm @ 700 nm, 10 nm @ 1400/2100 nm
Wavelength	350nm - 2500nm	350nm - 2500nm	350nm - 2500nm

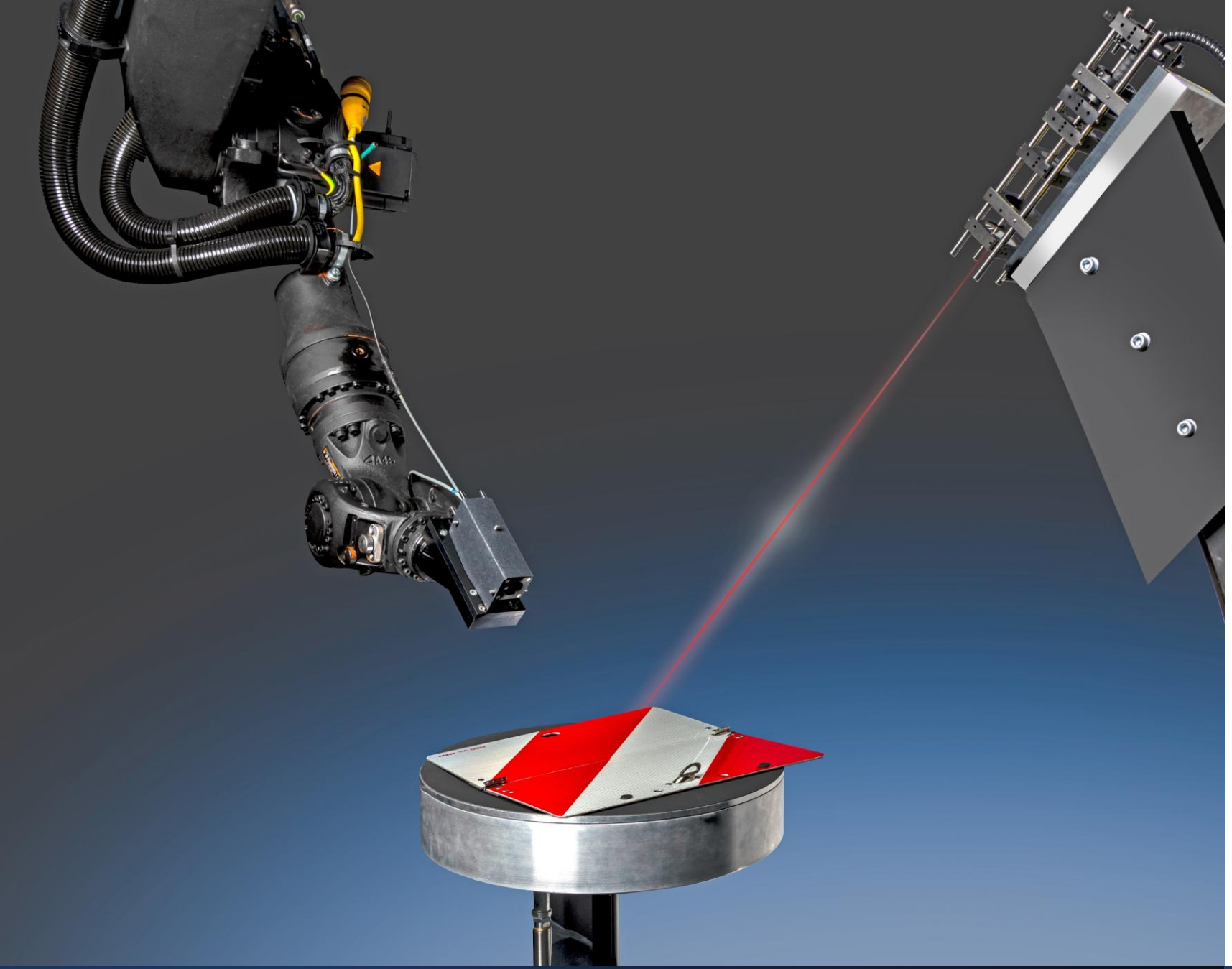
https://www.malvernpanalytical.com/en/products/product-range/asd-range/fieldspec-range?utm_source=google&utm_medium=cpc&utm_campaign=EN%20-%20ISEAA%20-%20Search&utm_term=field%20spectroradiometer&utm_content=EN%20-%20Product%20Category%20-%20Spectroradiometers&gad_source=1&gclid=CjwKCAiAlcyuBhBnEiwAOGZ2S25v9Z21wpXDTpq5EgroBiqY9F1Ut_eeJZK12Uvf_1VKe8GyqRAH7xoCXc0QAvD_BwE

Specs

Light Source:	Source Power Drift: 0.03% + 5mA Solid Angle: 2.4×10^{-5} sr
Detector:	Solid Angle: 0.00128 sr Signal Readout Resolution: 16 bit Detector Dynamic Range: 1:22,000 Electron Sensitivity: 1,900 photoelectrons/count Spectral Range: 386nm-711nm Spectral Resolution: 2.8nm
Mechanical System:	Rotation Stage 1 Range: 0°~360° Rotation Stage 1 Resolution: 0.01° Rotation Stage 2 Range: 0°~180° Rotation Stage 2 Resolution: 0.01° Rotation Stage 3 Range: 7°~180° Rotation Stage 3 Resolution: 0.013°
Test Sample:	Dimension of Sample Surface: 130mm×130mm Illumination Spot: 25mm diameter Uniformity of Illumination: within ±5% Maximum Angle of Reflection: ≥85°

Table 2 Reciprocity assessment on Spectralon with the Gonioreflectometer

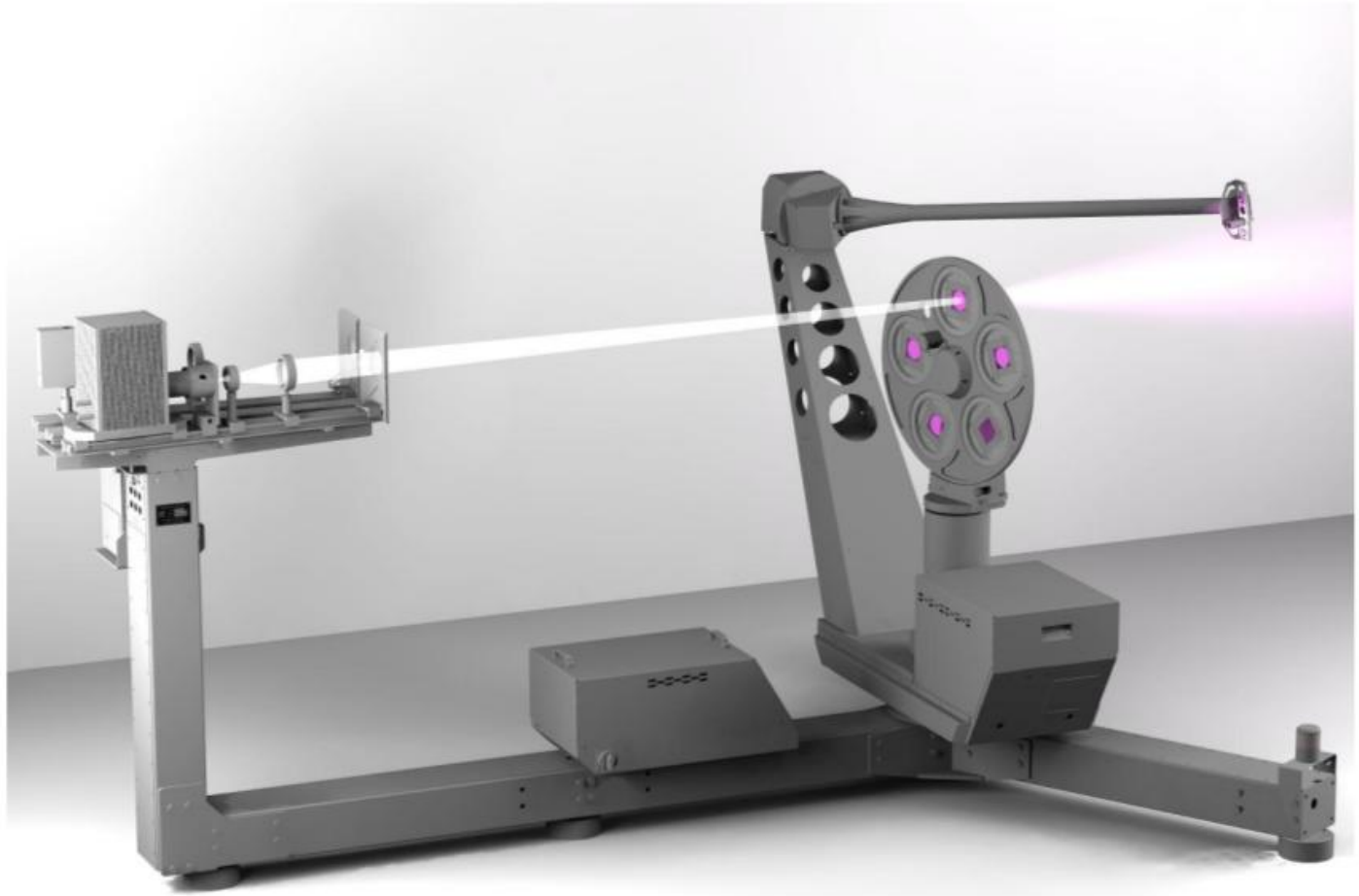
θ_i	θ_r	Signal Reading	Relative Error
40	80	1369	0.58%
80	40	1361	
46	74	1497	0.60%
74	46	1488	
50	70	1555	0.64%
70	50	1545	
54	66	1582	0.19%
66	54	1579	



Fraunhofer gonio-reflectometer

- https://www.iosb.fraunhofer.de/en/projects-and-products/sig_brdf_measurement/gonioreflektometer-messanlage.html

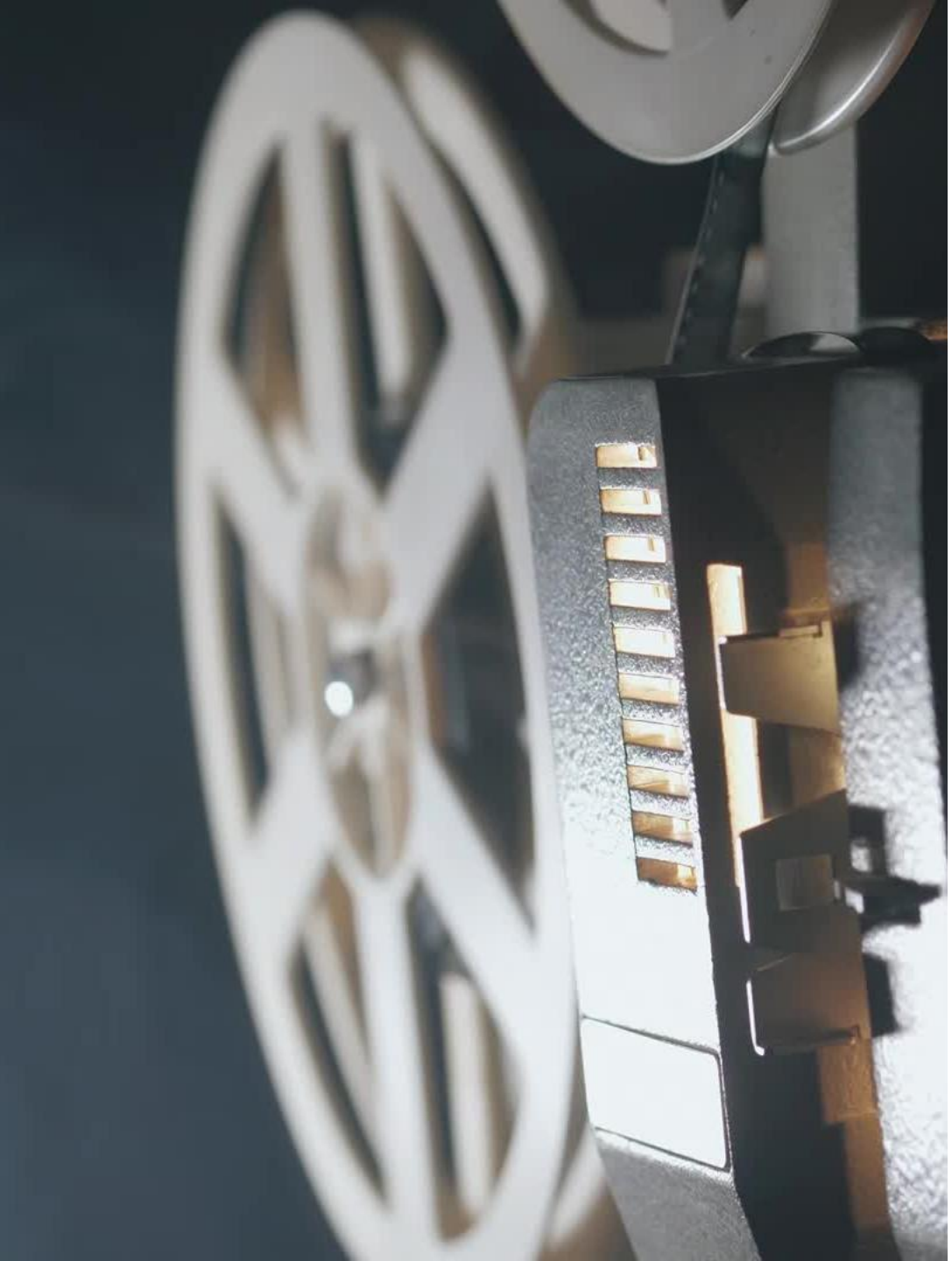
Gonio-Photometer for BSDF reflection & transmission measurements



pab gonio-photometer PG2 is specifically designed for fast and precise measurements of reflective and transmissive samples in research and commercial applications. Outstanding performance in speed and angular resolution is achieved with high performance servo drives, a rigid mechanical frame, optimized axis controls and industry leading software. It summarizes 20 years of experience in BSDF and light scattering measurements.

http://www.pab.eu/docs/PG2-flyer-en.pdf

Gonio-photometer in action



Keys of gonio-reflectometer setup

- Equipment
  - Sensor
  - Light source
  - Mechanical arms
  - Sample holder

- Configurations
  - Move light source and sensor but fix sample.
  - Move light source and sample but fix sensor.
  - Move everything !

Image-based measurement

- Extended use of imaging technology
- Capturing multiple samples at a time
- Cheaper equipment cost
- Leverage the use of material shape

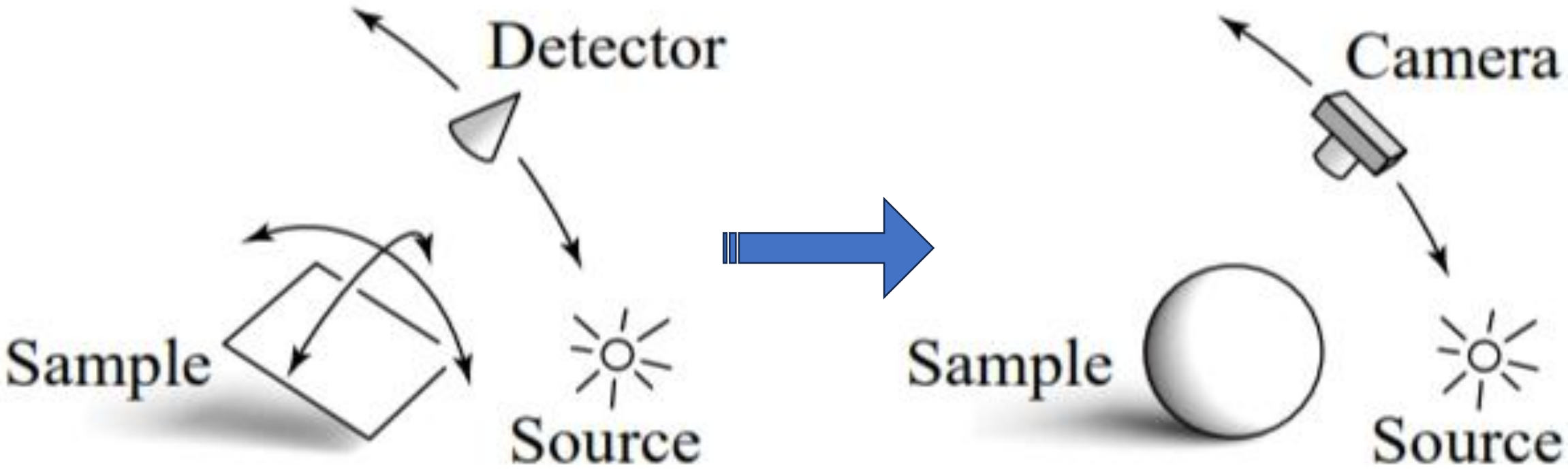
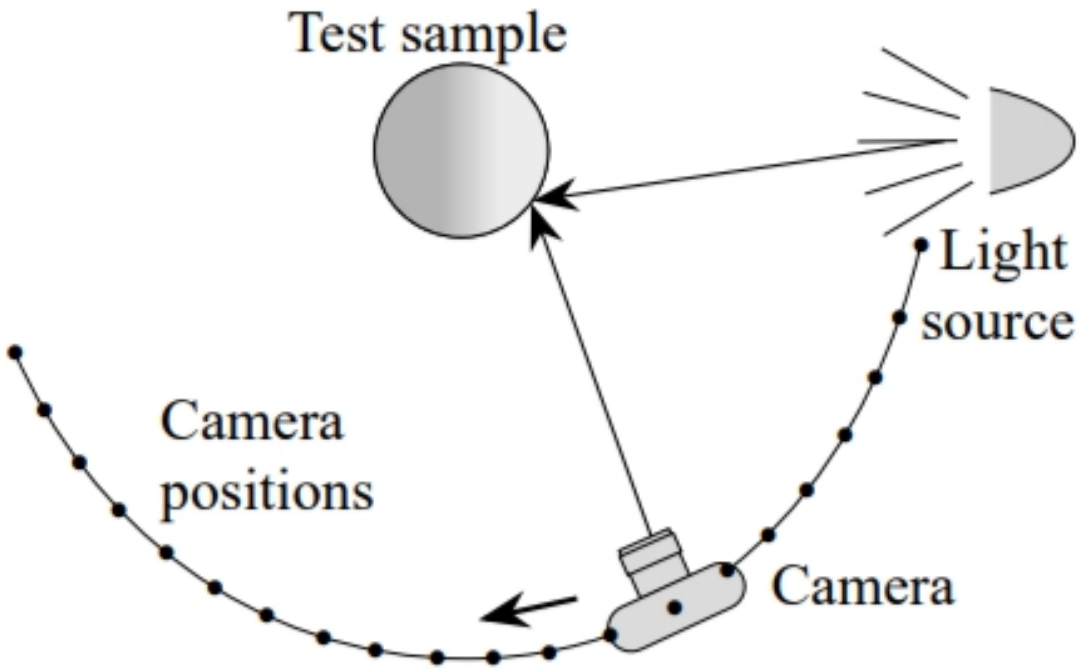
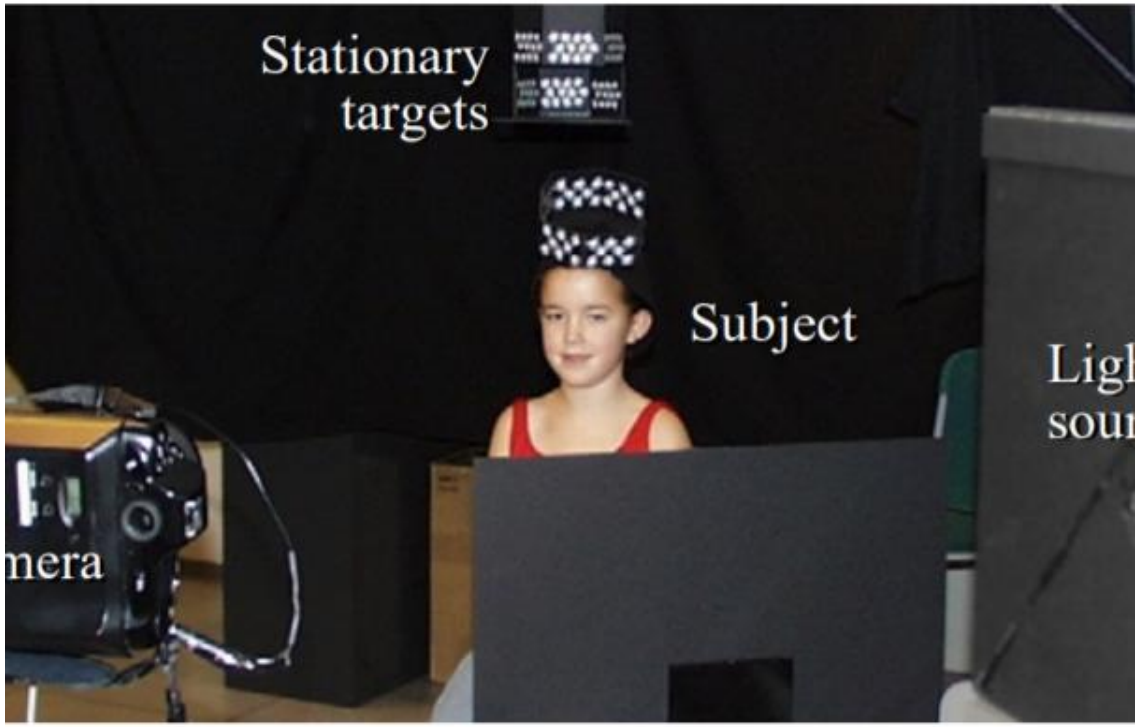


Image-based configuration

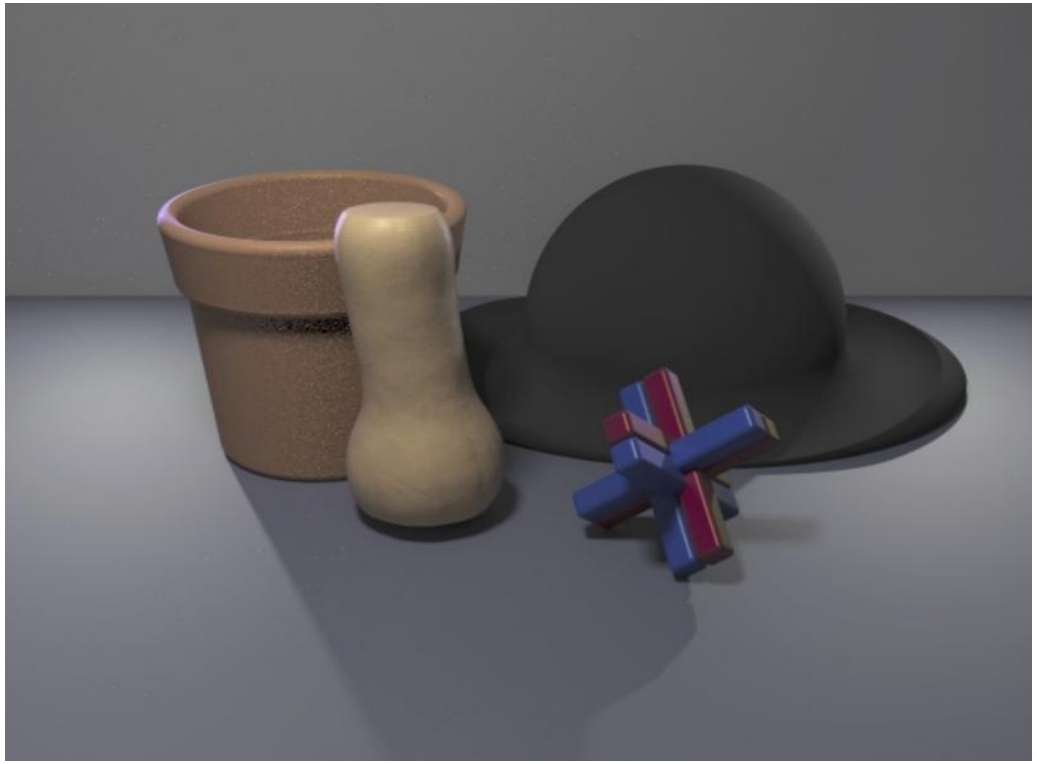
Marchner et al. proposed

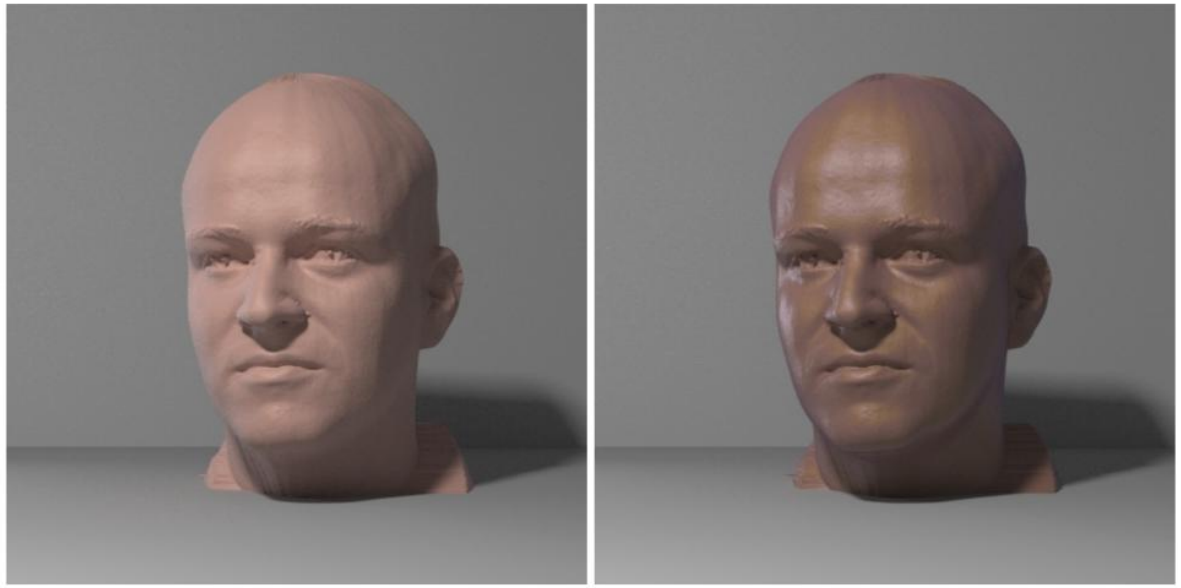




https://www.cs.cornell.edu/~srm/publications/EGRW99-brdf.pdf

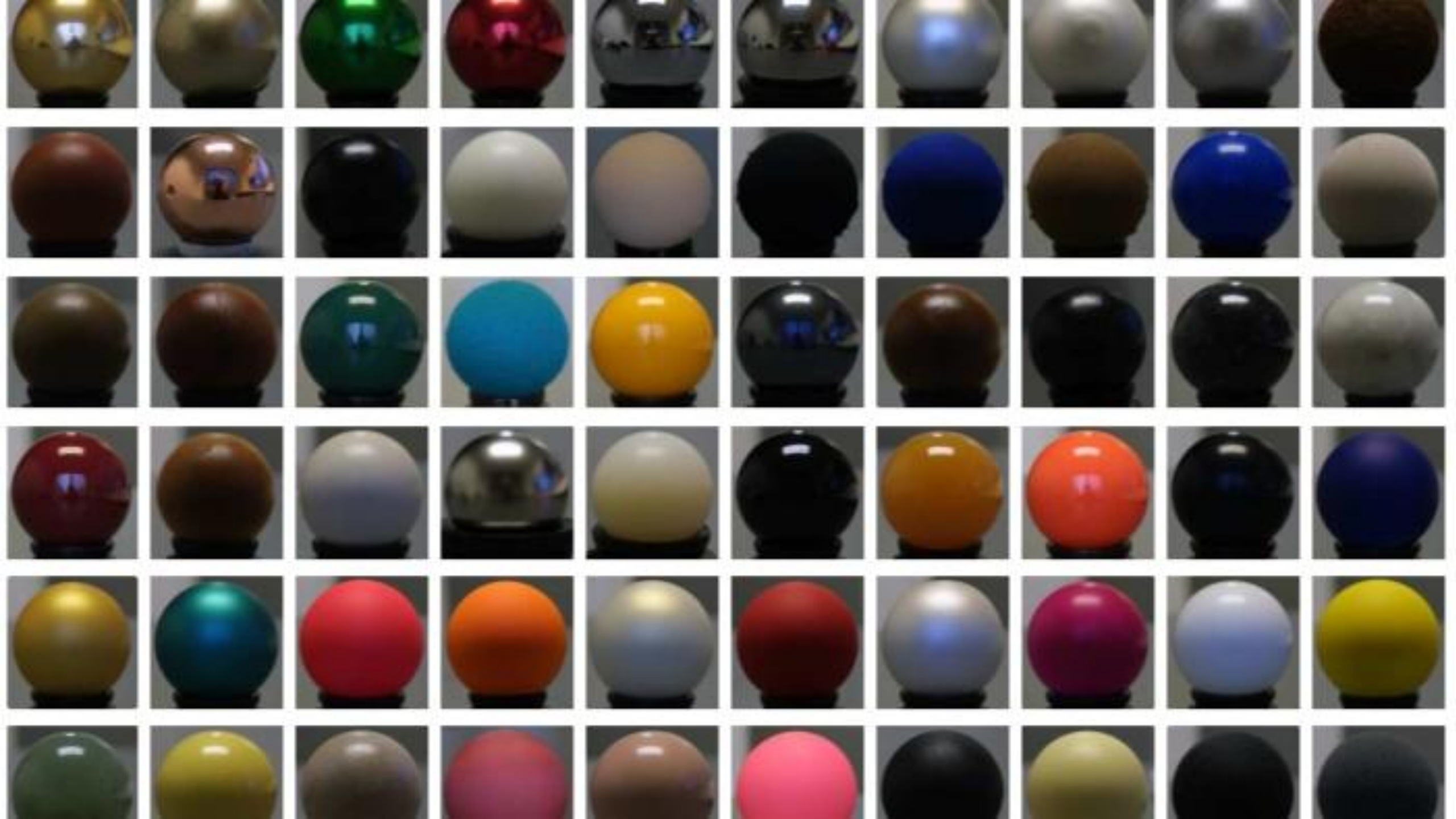
Results



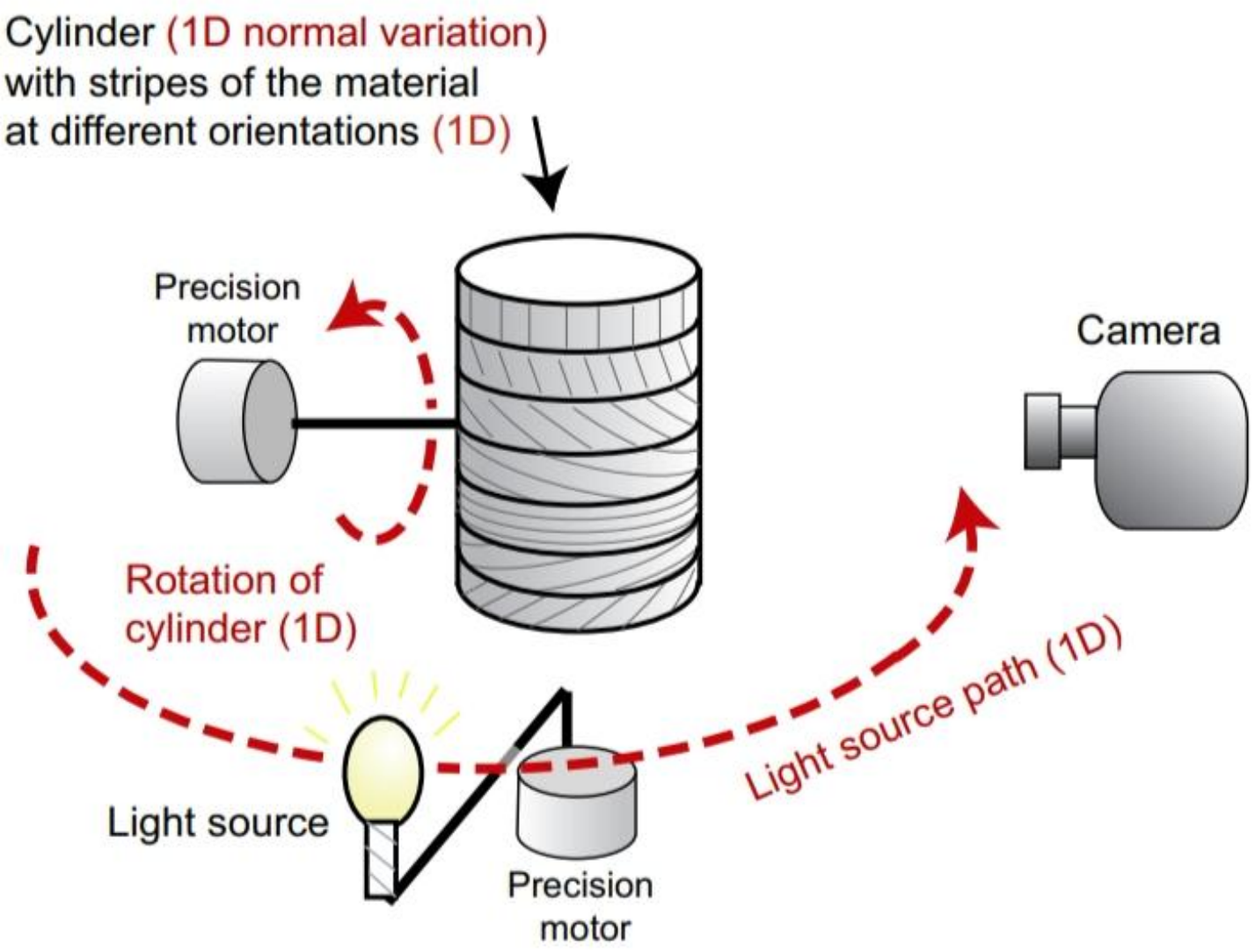


Matusik et al. proposed

https://www.merl.com/publications/docs/TR2003-83.pdf



An image-based anisotropic BRDF acquisition



https://people.csail.mit.edu/addy/research/ngan05_brdf_eval.pdf

BRDF Fitting

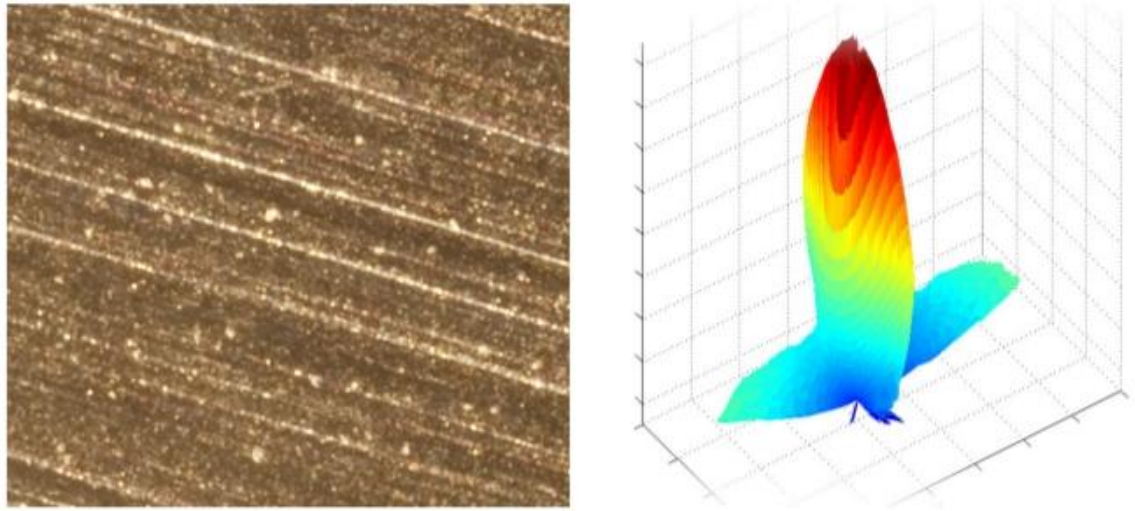


Figure 11: *Left: Brushed aluminium macro photograph. Right: deduced microfacet distribution (log plot).*

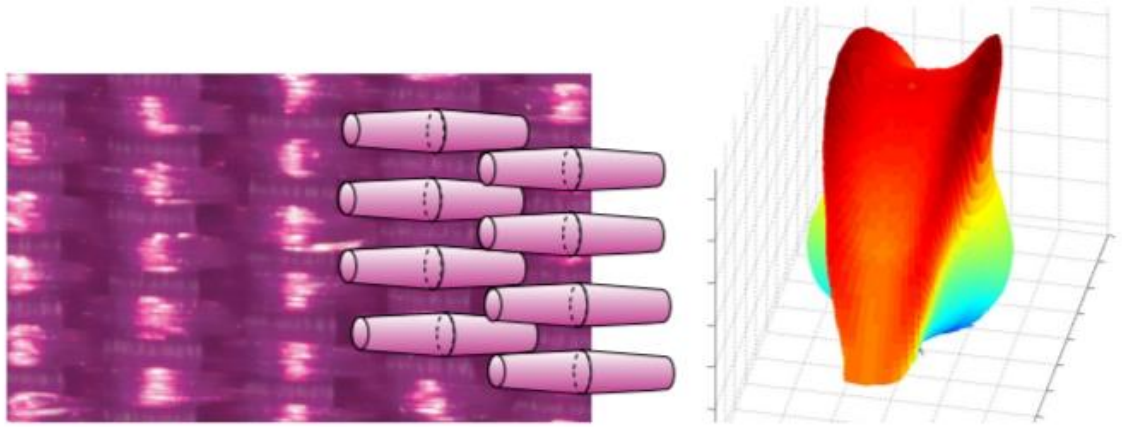


Figure 12: *Left: Purple satin macro photograph with sketched double cones. Right: deduced microfacet distribution (log plot).*

Microfacet BRDF fitting on measured samples

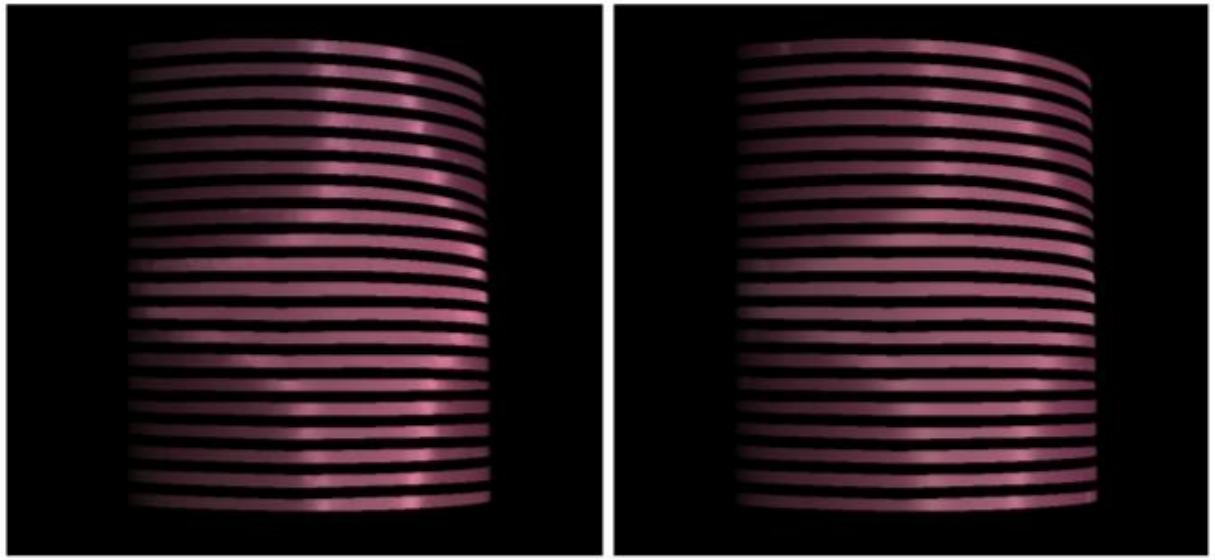
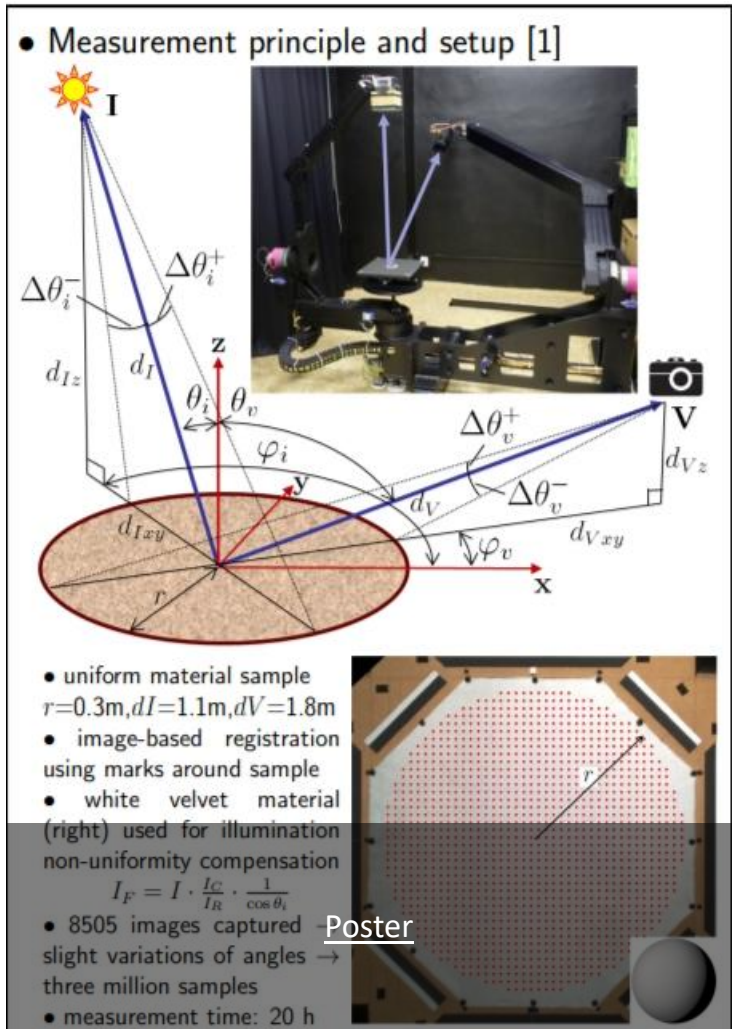
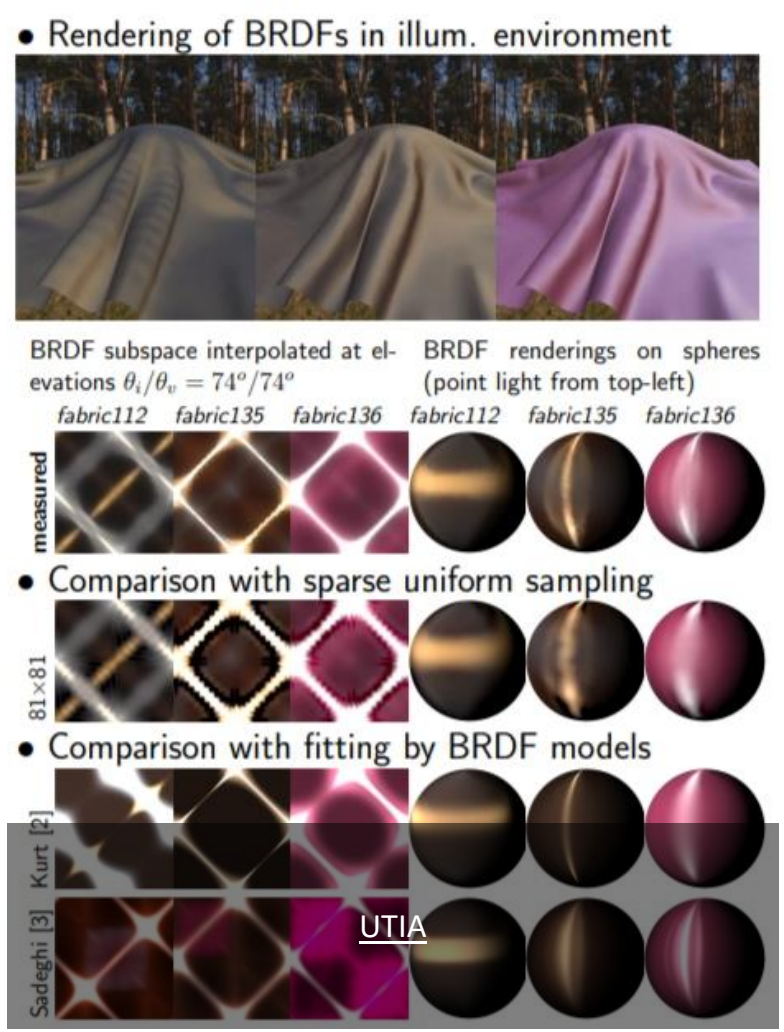
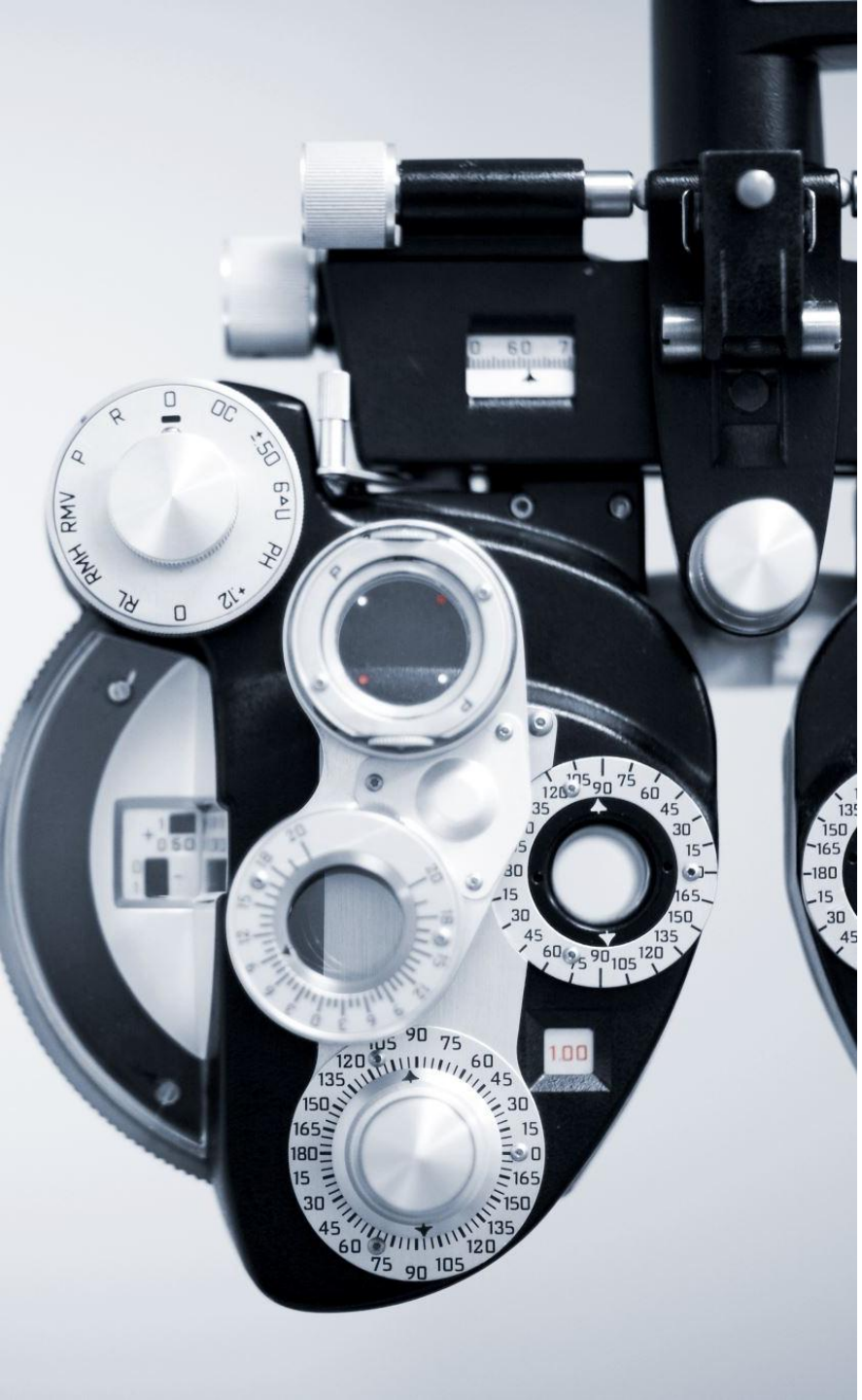


Figure 10: *Left: Purple satin - one input photograph. Right: - reconstruction using sampled microfacet distribution.*

Anisotropic BRDF acquisition - UTIA



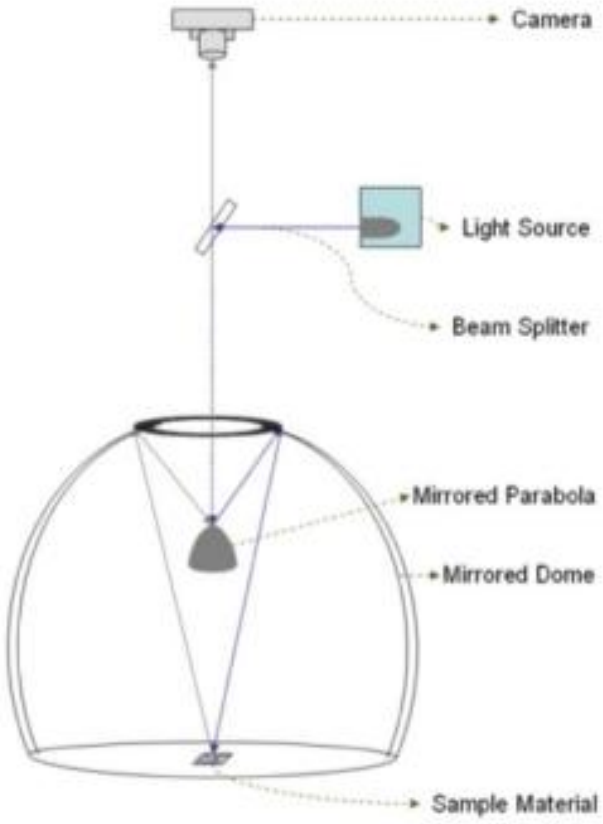


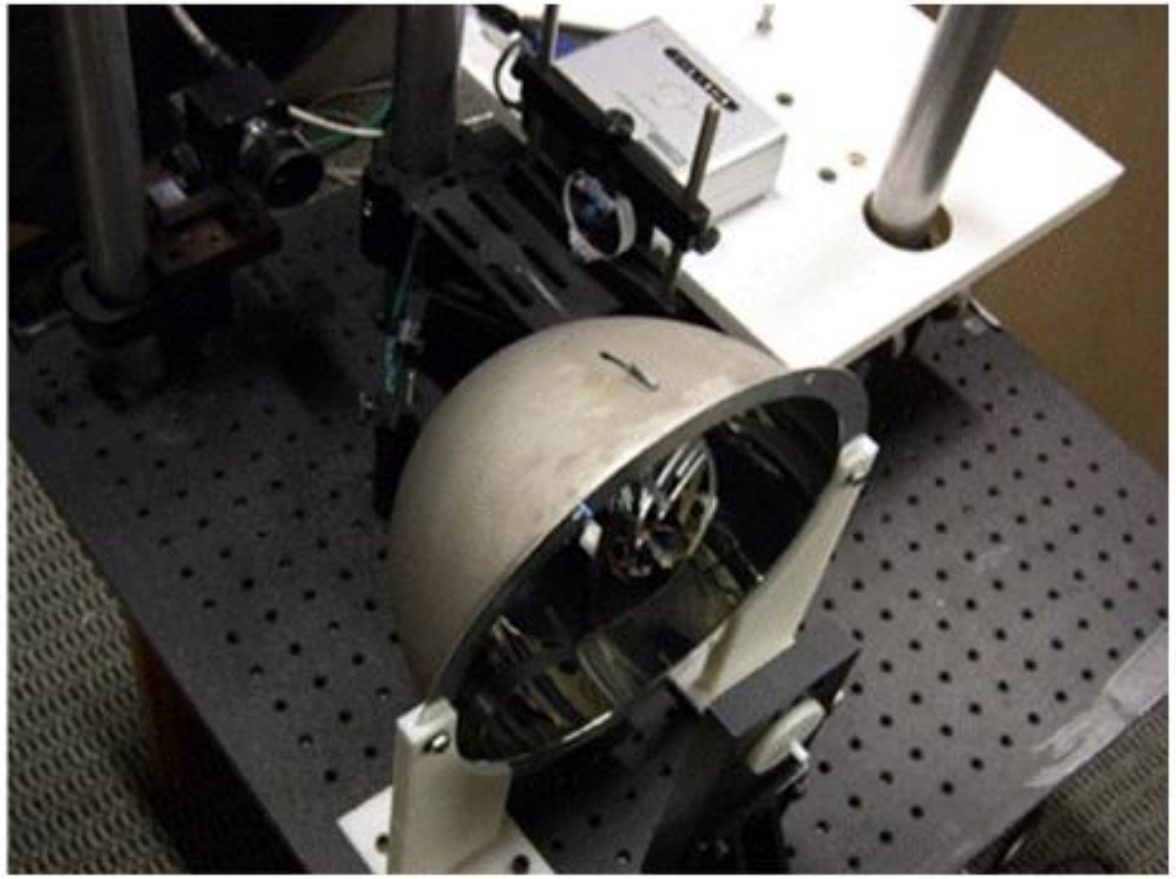


Catadioptric measurement setup

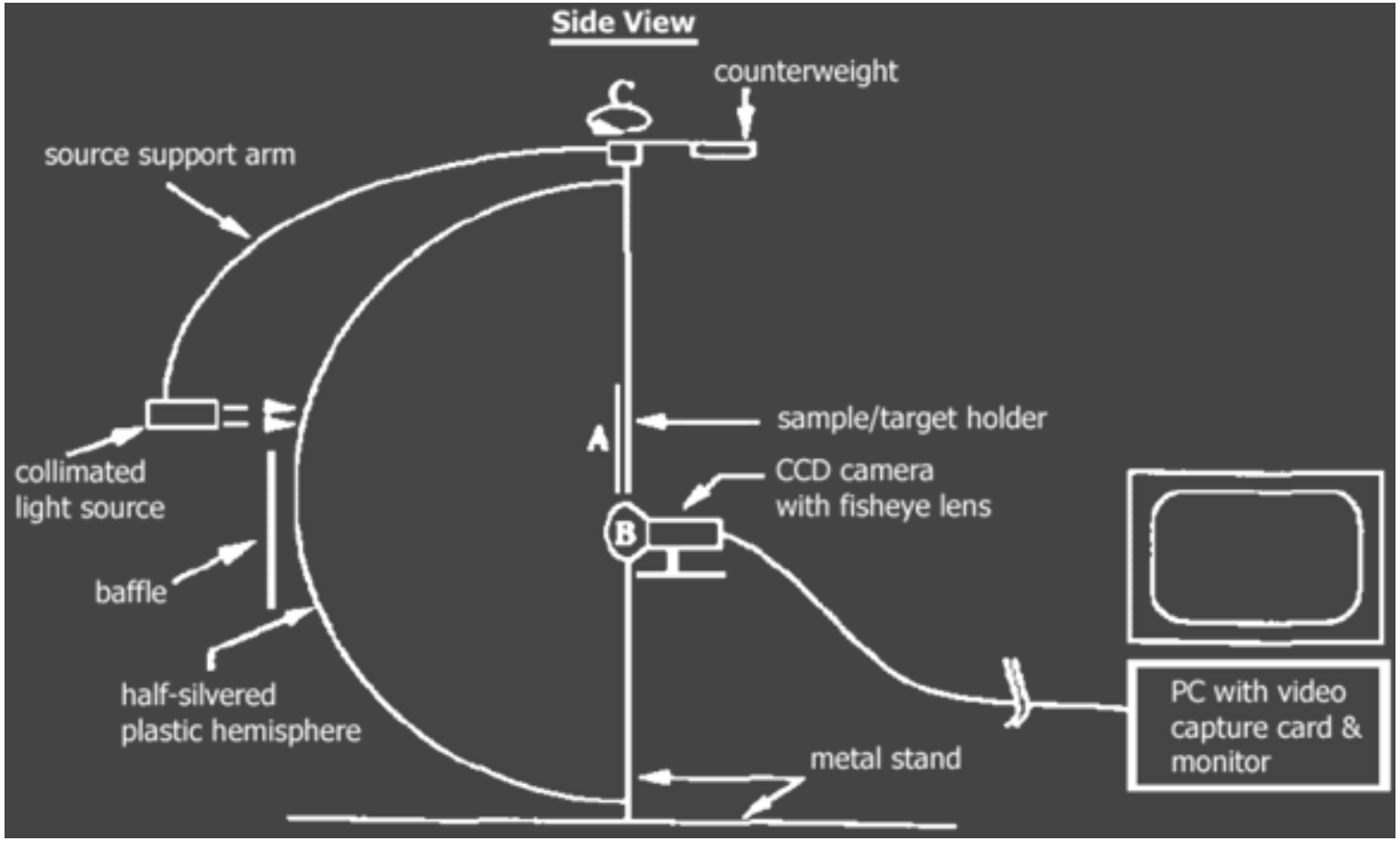
- Keys to setup
  - Catadioptric optical systems makes use of both reflected and refracted light, in order to reduce aberrations.
  - The resulting imaging setups are generally efficient image based BRDF acquisition devices, usually without any moving parts.
  - Invented reflection objects to handle various scattering situations.

Parabolic mirrors



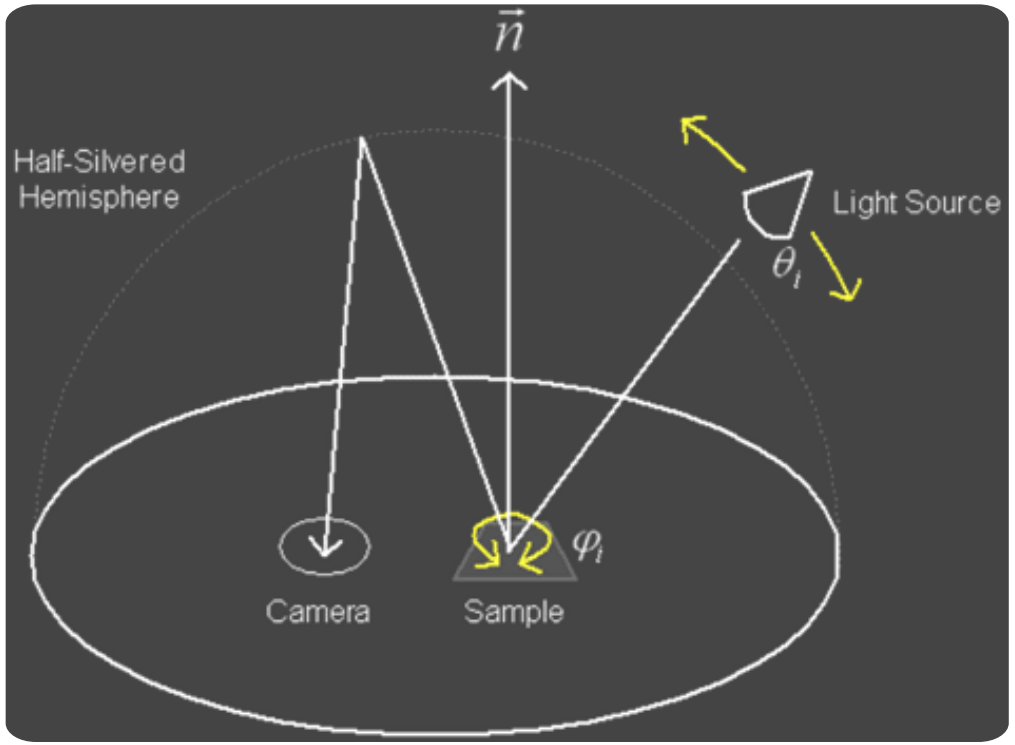


Hemispherical mirror



This device is designed to capture an entire hemisphere of reflected light at once. Capturing BRDF with this device is much faster than that of traditional gonioreflectometers.

Ward states that a complete BRDF can be recorded in a matter of minutes which includes the time for manually rotating the sample. In addition, it can capture anisotropic reflectance distribution.



- It employs a CCD camera fitted with a fish-eye lens to measure reflectance gathered by a half-silvered mirror from a target sample. The light source is collimated, meaning that it is designed so that the light rays are made parallel, and is setup on a arm outside the hemisphere. The hemisphere allows light through to the target sample. One benefit of this arrangement, that Ward states is that this setup allows for the measurement of retroreflection where the light is reflected back towards the light source.

How to obtain BRDF values from LDR images ?

Computing BRDF values from LDR images

- There are many steps in order to obtain BRDF values from captured images.
  - Steps to obtain radiance value from a HDR image
    - Camera calibration (done once)
    - Capturing LDR images with multi-exposure
    - Combining LDR images (at a position) to a HDR image
    - Compute radiance from a HDR image
  - Steps to obtain a BRDF value
    - Measure a target material
    - Measure a spectralon at the same locations
    - Compute a BRDF value from the measurements

HDR image from LDR images

```
% Loop over exposures
HDR = 0;
w_acc = 0;
for i = 1 : size(fnames,1)
    exp_t = sscanf(fnames(i).name, sprintf('img_%.3d.%.2d.%%d.png', a, i-1));

    % Load image
    I = double(imread(sprintf('%s/RAW/%s', path, fnames(i).name)));
    if(i == 1)
        HDR = zeros(size(I));
        w_acc = HDR;
    end
    % Load black level frame
    black = double(ReadPFM(sprintf('%s/black.%d.pfm', calib_path, exp_t)));
    % Radiometric calibration
    I = saturation * max((double(I) - black), 0) ./ (saturation - black);
    w = I;
    w(find(w > saturation)) = 0;
    HDR = HDR + w .* double(I) / exp_t;
    % Fuse exposures as weighted mean
    w_acc = w_acc + w;
end %  for i = 1 : size(fnames,1)
w_acc(find(w_acc == 0)) = 1;
HDR = HDR ./ w_acc;
```

Acknowledge : Saghi Hajisharif, Behnaz Kavoosi, Computer Graphics research group at LiU.

BRDF computation related to Spectralon

$$f_r(\theta_i, \theta_o, \phi_o) = \frac{I^{(M)}(\theta_i, \theta_o, \phi_o)}{I^{(S)}(\theta_i, \theta_o, \phi_o)} \qquad [\mathrm{sr}^{-1}]$$

Where:

$I^{(M)}$ is the intensity of the material. $[\,\cdot\,]$

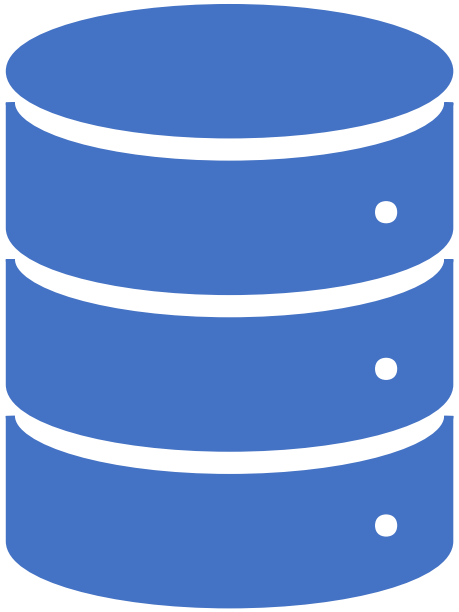
$I^{(S)}$ is the intensity of Spectrolon. $[\,\cdot\,]$

https://orbit.dtu.dk/files/128903442/phd435_Nielsen_JB_reduced3.pdf

Spectralon

SVBRDF / BTF capturing

- Very challenging as they (SVBRDF, BTF) are high-dimensional data.
- The combinations for dense sampling are far more than BRDF dense sampling.
- The storage size is huge.
- It takes time to capture a material.
- Bonn BTF dataset, Bonn SVBRDF dataset
- SVBRDF dataset
- UTIA BTF dataset



Material estimation

- ControlMat: Controlled Generative Approach to Material Capture
- MatFusion: A Generative Diffusion Model for SVBRDF Capture
- Single-Image SVBRDF Capture with a Rendering-Aware Deep Network
- Guided Fine-Tuning for Large-Scale Material Transfer